Root Security Audit & Remediation (root_audit.sh)

A- Application Overview and Objectives

The `root_audit.sh` utility is a security orchestration script for RHEL 9+ or Ubuntu 24.04+ environments. Its primary objective is to enforce a rigorous security posture for the `root` account, aligning with modern CIS and STIG compliance requirements.

The application serves three primary functions:

1. **Security Audit**: A non-intrusive, read-only assessment of the root account's current state (password status, lock state, SSH accessibility, and specific cryptographic hash algorithms like Yescrypt or SHA-512).
2. **Automated Remediation**: A deterministic process to harden the root account by enforcing complex 32-character passwords, locking the account (including re-locking after password updates), and disabling direct SSH access.
3. **Configuration Restoration**: A reliable rollback mechanism to restore system configurations from timestamped backups created during remediation.

B- Architecture and Design Choices

B1. Modular Functional Design

The script is architected as a collection of discrete, idempotent functions. This high-level modularity includes:

- **Service Isolation**: Dedicated functions for service management (e.g., `reload_ssh_service`).
- **Audit Orchestration**: Centralized `run_full_audit` for consistent state checking.
- **Security Logic**: Decoupled helpers for sudoers path discovery and group membership detection.
- **Testability**: Functions can be sourced and unit-tested in isolation without triggering the `main` execution flow.

B2. Operational Safety & Atomicity

Security tooling that modifies core system files like `/etc/shadow` or `/etc/sshd_config` must be fail-safe.

- **Atomic Edits**: Modifications to configuration files use a `mktemp` -> `sed` -> `mv` pattern. The `mv` command ensures an atomic replacement, preventing system files from being left in a corrupted state if the script is interrupted.
- **Automatic Backups**: Every file modification is preceded by a timestamped backup (`file.OBLIM.YYYYMMDDhhmmss`) with preserved attributes (`cp -p`).
- **Immutable Attribute Handling**: The script detects and handles the Linux `immutable` (`i`) attribute using `lsattr` and `chattr`, ensuring it can operate on systems hardened with filesystem-level locks.
- **SELinux Context Preservation**: The utility is SELinux-aware. It automatically executes `restorecon` after file modifications on RHEL-family systems to ensure that core configuration files retain their mandatory security labels. This logic is a safe no-op on AppArmor-based systems (Ubuntu).

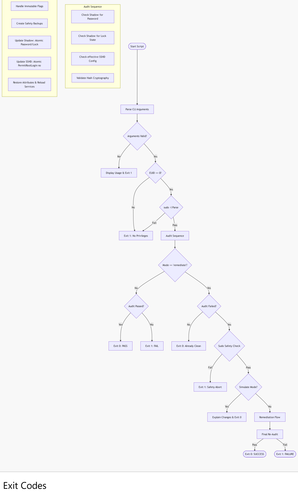
B3. Non-Interactive Orchestration

Designed for execution via automation platforms (Ansible, Puppet, Salt) or service accounts, the script is fully non-interactive. It relies on deterministic CLI arguments and returns standard exit codes (0 for success/pass, non-zero for failure).

B4. Sudo Safety "Kill-Switch"

A critical architectural safety feature is the `verify_sudo_users` function. It prevents administrative "brick" scenarios by aborting any destructive remediation (like locking root) if it cannot confirm the existence of at least one other non-root user with `sudo` privileges. To ensure total policy visibility, the script recursively scans the main `/etc/sudoers` file and all sub-policies within `/etc/sudoers.d/`.

C- Data Flow and Control Logic



D- Exit Codes

The utility returns deterministic exit codes to facilitate integration with CI/CD pipelines and automated orchestration platforms.

Code	Meaning	Context
`0`	**Success / Pass**	Audit passed, Remediation fully verified, or Restoration completed successfully.
`1`	**Failure / Error**	Audit failed, a specific Remediation step failed, a backup was missing during Restore, or privilege check failed.

*Note: In `remediate` and `restore` modes, the exit code reflects the **aggregate** success of all operations. If a single file fails to restore or a single hardening check fails final verification, the script will return `1`.*

E- Dependencies

The script utilizes standard Linux core utilities available on all RHEL/Ubuntu minimal installations. No third-party modules or libraries are required.

- **Shell**: `bash` (4.0+)
- **Core Utils**: `grep`, `sed`, `awk`, `cut`, `tr`, `find`, `mktemp`, `cp`, `mv`, `date`,
- **User Management**: `passwd`, `chpasswd`, `getent`,
- **System Management**: `systemctl` (for SSHD reload), `sudo`, `script` (for TTY allocation).
- **Security**: `lsattr`, `chattr` (for immutable flag handling), `sshd` (for `-T` config auditing).

F- Command Line Arguments

Argument	Type	Description
AUDIT Mode		
`--mode audit`	String	Read-only security check (default).
REMEDIATE Mode		
`--mode remediate`	String	Fix security issues (if missing only).
`--password <PASS>`	String	Provide a 32-char password. **Security Note**: Prefer using the `ROOT_PASSWORD` environment variable to avoid process list visibility.
`--generate`	Flag	Automatically generates a secure 32-char password.
RESTORE Mode		
`--mode restore`	String	Restore configuration from backup.
`--suffix <SUFFIX>`	String	Suffix of backup to restore (e.g., `20260423160441`).
`--list-backups`	Flag	Lists available backups in a tabular format (Suffix, Date, Files, Size).
`--latest`	Flag	Automatically selects the most recent backup for restoration.
`--oldest`	Flag	Automatically selects the oldest available backup for restoration.
COMMON		
`--simulate`	Flag	Explain changes without applying them (Remediate/Restore mode only).
`--json`	Flag	Enables CI/CD friendly JSON output on STDOUT.
`--log <file>`	String	Log all messages to a specific file (ANSI codes stripped).

F1. Configuration via Environment Variables

For advanced orchestration, the following variables can be set to override defaults:

- `SHADOW_FILE`: Path to shadow file (default: `/etc/shadow`)
- `SSH_CONFIG`: Path to sshd_config (default: `/etc/ssh/sshd_config`)
- `SUDOERS_FILE`: Path to main sudoers (default: `/etc/sudoers`)
- `SUDOERS_DIR`: Path to sudoers.d directory (default: `/etc/sudoers.d`)
- `MIN_PASS_LEN`: Minimum password length policy (default: 32)
- `SUFFIX_DELIM`: Delimiter for backup files (default: `.`)

G- Usage Examples

G1. Standard Audit

Performs a read-only security check.

```
sudo ./root_audit.sh --mode audit
```

Note: Executing the script without arguments will display the usage information.

G2. Automated Remediation with Generation

Generates a 32-character password, locks the account, and hardens SSH.

```
sudo ./root_audit.sh --mode remediate --generate
```

G3. Manual Remediation

Uses a pre-defined password that meets the 32-character policy.

```
sudo ./root_audit.sh --mode remediate --password
'v@ryStr0ngP@ssw0rd_32_Chars_Long!!'
```

G4. Configuration Restoration (Manual)

Roll back changes using a specific timestamped backup suffix.

```
sudo ./root_audit.sh --mode restore --suffix 20260423160441
```

G5. Backup Discovery

List all available timestamped backups, their creation dates, and their file scope.

```
sudo ./root_audit.sh --mode restore --list-backups
```

G6. Automated Restoration

Quickly roll back to the most recent or oldest available configuration state.

```
# Restore the latest backup
sudo ./root_audit.sh --mode restore --latest

# Restore the oldest backup
sudo ./root_audit.sh --mode restore --oldest
```

G7. Integration via Environment Overrides

Used for testing or non-standard filesystem layouts.

```
SSH_CONFIG="/custom/sshd_config" sudo ./root_audit.sh --mode audit
```

G8. Secure Remediation (Environment Variable)

To avoid sensitive data appearing in process lists (`ps`) or shell history, use the `ROOT_PASSWORD` environment variable.

```
# Pass password securely via environment
export ROOT_PASSWORD='v@ryStr0ngP@ssw0rd_32_Chars_Long!!'
sudo -E ./root_audit.sh --mode remediate
```

Note: The `-E` flag for `sudo` is used to preserve the environment variable.

H- Unit Tests

The `root_audit_test.sh` suite provides comprehensive validation of the application's logic, safety mechanisms, and error handling. It is designed to run in a fully isolated environment without modifying the host system.

H1. Test Execution

To ensure the sandboxed environment can correctly simulate protected system states (like immutable flags), the test suite **must be run with sudo**:

```
sudo ./root_audit_test.sh
```

If you encounter the error `sudo: sorry, you must have a tty to run sudo` (common in CI/CD or automated environments), use the `script` utility to allocate a pseudo-terminal:

```
script -q -c "sudo ./root_audit_test.sh" /dev/null
```

Note: The test suite creates a temporary workspace in `$TMPDIR/unitests/$uuid` and performs an automatic cleanup upon completion.

H2. Test Coverage Scenarios

The suite covers **72 distinct validation points**, categorized as follows:

Category	Scenarios Covered
Password Policy	Exhaustive validation of length and complexity (uppercase, lowercase, digits, special characters) including rejection paths.
Attribute Handling	Detection, temporary removal, and restoration of the `immutable` (i) flag during file operations.
File Safety	Verification of timestamped backups (using `$UFFIX_DELIM`) and atomic state transitions.
Audit Engine	Logic validation for detecting unlocked accounts, weak cryptographic hashes (MD5), and insecure SSH configurations.
SELinux Awareness	Automatically restores security contexts using `restorecon` if SELinux is active.
Sudo Safety Kill-Switch	Multi-path discovery simulation for administrators (groups and explicit entries) to prevent administrative lockout.
CLI Orchestration	Comprehensive exit code validation for all modes, no-argument guards, and environment variable (`ROOT_PASSWORD`) injection.
JSON Mode	Verification of schema integrity and machine-readable output for CI/CD integration.
Restore Logic	Validation of rollback integrity, file content restoration, and missing backup handling.
Backup Discovery	Automated identification of latest/oldest backups and tabular reporting of backup scope (shadow vs sshd_config).
Privilege Logic	Strict enforcement of `EUID=0` requirements for both successful execution and graceful failure.

I- Remote Execution via SSH

When executing `root_audit.sh` remotely via SSH, you can pass the `ROOT_PASSWORD` variable securely using the following patterns.

I1. Direct Command Prefix (Recommended)

This is the most common method. By prefixing the command with the variable assignment and using `sudo -E`, the variable is passed to the script's environment.

```
ssh -t user@remote-host "export ROOT_PASSWORD='your-secure-password'; sudo -E
/path/to/root_audit.sh --mode remediate"
```

Note: The `-t` flag is often required for `sudo` prompts over SSH.

I2. Piped Execution (No script on host)

If the script is not already present on the remote host, you can stream it over SSH while passing the environment.

```
cat root_audit.sh | ssh user@remote-host "export ROOT_PASSWORD='your-secure-
password'; sudo -E bash -s -- --mode remediate"
```

I3. SSH Configuration (SendEnv)

If your organization allows it, you can configure SSH to pass the environment variable automatically.

1. **Remote Host**: Add `AcceptEnv ROOT_PASSWORD` to `/etc/ssh/sshd_config`.
2. **Local Host**: Use `SendEnv` in your SSH command.

```
export ROOT_PASSWORD='your-secure-password'
ssh -o SendEnv=ROOT_PASSWORD user@remote-host "sudo -E /path/to/root_audit.sh --
mode remediate"
```

J- Simulation Mode

The `--simulate` flag allows you to dry-run the remediation process. It performs a full audit and sudo safety check, then explains every step it *would* take without modifying the system.

J1. Example Simulation

```
sudo ./root_audit.sh --mode remediate --generate --simulate
```

J2. Simulation Output Breakdown

- **Password**: Explains that a 32-character password would be enforced.
- **Locking**: Explains that the account would be locked in the shadow file.
- **SSH Hardening**: Details the atomic modification of `sshd_config` and the service reload.
- **Safety**: Confirms that backups and immutable attribute handling would be performed.

K- JSON Mode for CI/CD

The `--json` flag enables machine-readable output on `STDOUT`. This is ideal for integration with automation tools, monitoring systems, and security orchestrators.

K1. Standard JSON Audit

```
sudo ./root_audit.sh --mode audit --json
```

K2. JSON Schema

The output is a structured object containing:

- `timestamp`: ISO-8601 execution time.
- `target_user`: Target account name.
- `mode`: Execution mode (`audit` or `remediate`).
- `immutable`: Boolean indicating if `chattr +i` was detected on scoped files.
- `status`: `success` or `failure`.
- `audit`: Granular results of all security checks (Audit mode only).
- `remediation`: Summary of actions taken (Remediate mode only).
- `restore`: Results of the configuration rollback (Restore mode only).

`jq` Support: If `jq` is installed, the output will be pretty-printed. If not, the script falls back to raw JSON strings. Standard log messages are preserved on `STDERR`.

K3. Simulation with JSON Output

Combine flags to preview remediation steps in a machine-readable format without making system changes.

```
sudo ./root_audit.sh --mode remediate --generate --simulate --json
```

This is the recommended pattern for validating remediation logic in CI/CD pipelines before production deployment.

K4. JSON Output Examples

Audit Mode

```
{
  "timestamp": "2026-04-23T14:47:54-05:00",
  "target_user": "root",
  "mode": "audit",
  "immutable": false,
  "simulate": false,
  "status": "failure",
  "audit": {
    "password_exists": "true",
    "account_locked": "false",
    "ssh_disabled": "false",
    "hash_strong": "true",
    "sudo_user_count": 2,
    "sudo_user_list": "builder,ubuntu"
  }
}
```

Remediate Mode

```
{
  "timestamp": "2026-04-23T14:50:12-05:00",
  "target_user": "root",
  "mode": "remediate",
  "immutable": true,
  "simulate": false,
  "status": "success",
  "remediation": {
    "password_updated": true,
    "account_locked": true,
    "ssh_hardened": true,
    "backup_created": true,
    "sudo_safety_passed": "true"
  }
}
```

Restore Mode

```
{
  "timestamp": "2026-04-23T16:15:00-05:00",
  "target_user": "root",
  "mode": "restore",
  "immutable": false,
  "simulate": false,
  "status": "success",
  "restore": {
    "shadow_restored": true,
    "ssh_restored": true
  }
}
```